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Questions

Q1.

* Compare and contrast the reactions of propanal and propanone with **one** oxidising agent, **one** reducing agent and 2,4-dinitrophenylhydrazine.

In your answer include any relevant observations for the reactions you discuss and equations for any reactions classified as oxidation, using [O] for the oxygen from the oxidising agent.

(Total for question = 6 marks)

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Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The following methods can be used to distinguish between pairs of organic compounds with **no** further tests.

A warm each compound with Fehling's or Benedict's solution

B warm each compound with acidified potassium dichromate(VI) solution

C add 2,4-dinitrophenylhydrazine (Brady's reagent) to each compound

D add a few drops of each compound, drop by drop, to water

(a) Which test would distinguish between 2-methylpropan-2-ol, $(CH_3)_3COH$, and butanone, $CH_3CH_2COCH_3$?

(1)

- ☐ **A**
- ☐ **B**
- ☐ **C**
- ☐ **D**

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(b) Which test would distinguish between 2-methylpropan-2-ol, $(CH_3)_3COH$, and butan-1-ol, $CH_3CH_2CH_2CH_2OH$?

(1)

- ☐ **A**
- ☐ **B**
- ☐ **C**
- ☐ **D**

(c) Which test could **not** be used to distinguish between 2-methylpropan-2-ol, $(CH_3)_3COH$, and butanal, $CH_3CH_2CH_2CHO$?

(1)

- ☐ **A**
- ☐ **B**

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☒ **C**
☒ **D**

(Total for question = 3 marks)

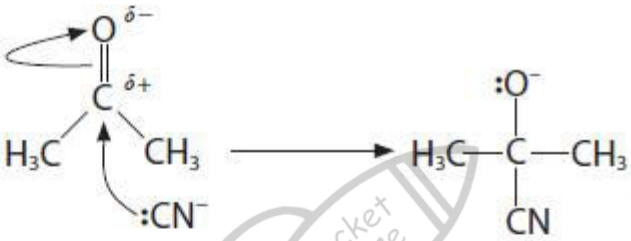
Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

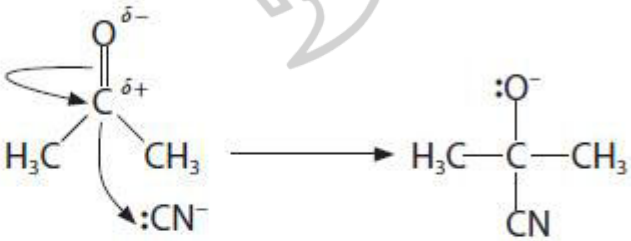
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Which diagram shows the first step of the mechanism for the nucleophilic addition of hydrogen cyanide to propanone in the presence of potassium cyanide?

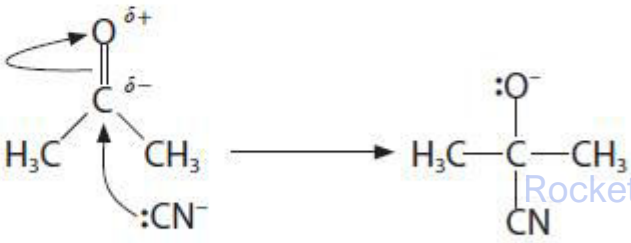
☒ **A**



☒ **B**

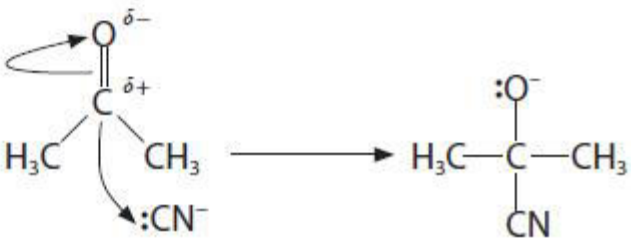


☒ **C**



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☒ **D**



Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

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Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When propanone reacts with iodine in the presence of sodium hydroxide, the precipitate formed has the formula

- ☒ **A** CH_3I
- ☒ **B** CHI_3
- ☒ **C** CH_3COCH_2I
- ☒ **D** CH_3COCI_3

(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An unknown aldehyde may be identified by measuring the melting temperature of the purified precipitate formed in its reaction with

- ☒ **A** 2,4-dinitrophenylhydrazine
- ☒ **B** Fehling's solution
- ☒ **C** potassium dichromate and sulfuric acid
- ☒ **D** Tollens' reagent

(Total for question = 1 mark)

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Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Ethanal and propane have the same molar mass but ethanal has a much higher boiling temperature.

Ethanal is fully miscible in water but propane is almost insoluble.

Which intermolecular forces of ethanal are mainly responsible for the differences in these properties?

	Higher boiling temperature	Greater solubility in water
☐ A	hydrogen bonds	hydrogen bonds
☐ B	permanent dipole forces	permanent dipole forces
☐ C	hydrogen bonds	permanent dipole forces
☐ D	permanent dipole forces	hydrogen bonds

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(Total for question = 1 mark)

Q7.

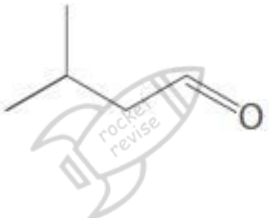
Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

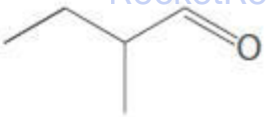
A compound **X**, with molecular formula $C_5H_{10}O$, gave an orange precipitate with 2,4-dinitrophenylhydrazine.

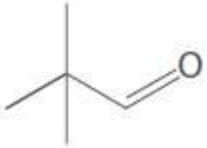
X gave a silver mirror with Tollens' reagent.

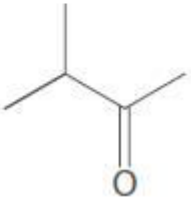
Which of these could **not** be **X**?

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☐ A 

☐ B 

☐ C 

☐ D 

(Total for question = 1 mark)

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Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which alcohol could **not** be formed by the reduction of an aldehyde or a ketone?

- ☒ **A** 2-methylpentan-1-ol
- ☒ **B** 3-methylpentan-2-ol
- ☒ **C** 3-methylpentan-3-ol
- ☒ **D** 2-methylpentan-3-ol

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(Total for question = 1 mark)

Q9.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these reactions does **not** result in the formation of a racemic mixture?

- ☒ **A** but-2-ene with HCl
- ☒ **B** but-1-ene with HCl
- ☒ **C** propanal with HCN
- ☒ **D** propanone with HCN

(Total for question = 1 mark)

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Q10.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Propanone reacts with iodine in both acidic and alkaline conditions.

(a) The products of the reactions under the stated conditions include

(1)

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	Acidic conditions	Alkaline conditions
☒ **A**	CH_3I	CH_3COO^-
☒ **B**	CH_3COCI_3	CH_3COO^-
☒ **C**	CH_3COCH_2I	CH_3I
☒ **D**	CH_3COCH_2I	CHI_3

(b) The rate equation for the reaction between propanone and iodine in acidic conditions is

$$\text{rate} = k[H^+][CH_3COCH_3]$$

The reaction was carried out at two different pH values, all other conditions remaining unchanged.
In the first reaction pH = 2.0
In the second reaction the rate was found to be ⅓ of the original value.
What was the pH of the second reaction, to 1 decimal place?

(1)

☒ **A** 0.7
☒ **B** 1.5
☒ **C** 2.5
☒ **D** 2.7

(Total for question = 2 marks)

Q11.

Propanal reacts very slowly with HCN at 298 K. To increase the rate of reaction potassium cyanide, KCN, is added.

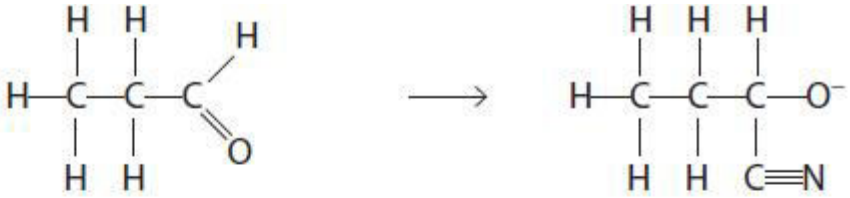
(a)(i) Complete the mechanism for this reaction by adding curly arrows, and relevant lone pairs and dipoles to Step **1** and Step **2**.

(4)

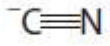
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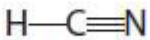
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Step 1



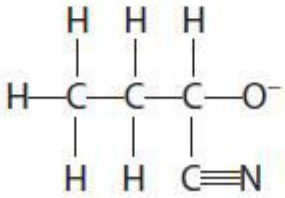
intermediate





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Step 2



products

intermediate

(ii) Explain why the reaction between propanal and HCN in the absence of KCN is very slow, referring to the value of K_a. No calculation is required.
[For HCN, $K_a = 4.9 \times 10^{-10}$ mol dm^{-3}]

(2)

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(iii) KCN is a **homogeneous catalyst** in this reaction.
Justify this description by referring to the mechanism.

(2)

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(b) The organic products of this reaction are enantiomers.

Draw the three-dimensional structures of these enantiomers.

(2)

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(Total for question = 10 marks)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound forms a precipitate with Fehling's or Benedict's solutions **and** with iodine in the presence of alkali?

- ☒ **A** methanal
- ☒ **B** ethanal
- ☒ **C** propanone
- ☒ **D** butanone

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(Total for question = 1 mark)

Q13.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many structural isomers with the molecular formula $C_5H_{10}O$ react with Tollens' reagent?

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☒ **A** 2
☒ **B** 3
☒ **C** 4
☒ **D** 5

(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Propanal has a much higher boiling temperature than butane even though both compounds have the same molar mass.

The main reason for the difference in boiling temperature is that in the liquid state

☒ **A** propanal forms much stronger London forces than butane
☒ **B** propanal forms permanent dipole-permanent dipole forces
☒ **C** propanal forms hydrogen bonds
☒ **D** propanal forms hydrogen bonds and permanent dipole-permanent dipole forces

(Total for question = 1 mark)

Q15.

Hexane-2,5-dione, $CH_3COCH_2CH_2COCH_3$, is a toxic compound formed in the human body if hexane is consumed.

(a) Complete the table for hexane-2,5-dione.

Name the organic product formed if a reaction takes place.

(2)

Reagent and conditions	Reaction (✓ / ✗)	Name of organic product (if formed)
refluxed with excess acidified potassium dichromate(VI)		
excess lithium tetrahydridoaluminate(III) in dry ether		

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(b) State the observation when hexane-2,5-dione reacts with iodine in the presence of alkali.

(1)

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(c) Hexane-2,5-dione reacts with **excess** hydrogen cyanide, HCN, in the presence of potassium cyanide, KCN.

(i) Name the type and mechanism of this reaction.

(1)

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(ii) Draw the structure of the product.

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(1)

(d) (i) Give the observation when 2,4-dinitrophenylhydrazine (Brady's reagent) reacts with hexane-2,5-dione.

(1)

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(ii) Describe, in outline, how the product of this reaction may be used to confirm the identity of hexane-2,5-dione. Experimental details are not required.

(2)

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(Total for question = 8 marks)

Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Samples of ethanal and propanone are added to separate test tubes containing iodine dissolved in aqueous potassium hydroxide and the mixtures are heated gently.

A pale yellow precipitate will be formed with

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- **A** ethanal and propanone
- **B** ethanal only
- **C** propanone only
- **D** neither ethanal nor propanone

(Total for question = 1 mark)

Q17.

Which compound gives a pale yellow precipitate on warming with a solution of iodine and sodium hydroxide?

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(1)

- **A** CH_3OH
- **B** $CH_3CH_2CH_2CHO$
- **C** $CH_3CH_2COCH_3$
- **D** $CH_3CH_2COCH_2CH_3$

(Total for question = 1 mark)

Q18.

This question is about carbonyl compounds such as ethanal, CH_3CHO.

(a) Ethanal reacts with the weak acid hydrogen cyanide to form a racemic mixture of hydroxynitriles.

$$CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$$

A small amount of sodium hydroxide solution, NaOH(aq), is also added to the reaction mixture.

(i) Complete the mechanism for the reaction.

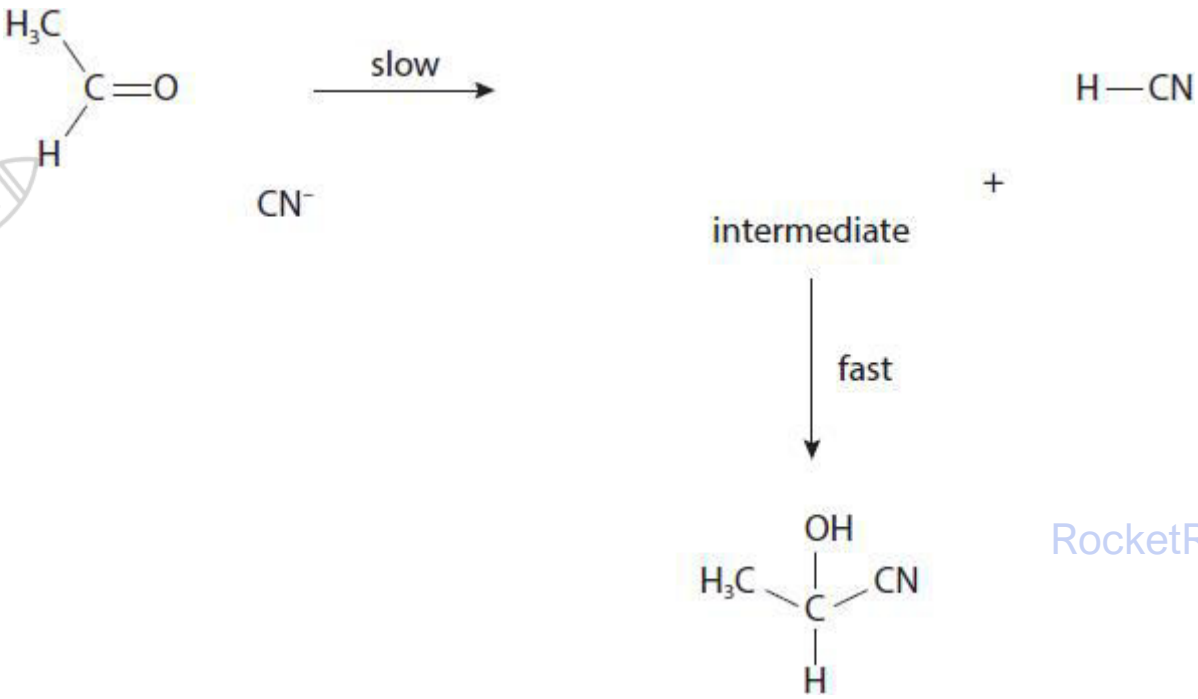
Include curly arrows, the structure of the intermediate, and any relevant lone pairs and dipoles.

(4)

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(ii) Explain why a small amount of sodium hydroxide solution is added to the reaction mixture, by considering the nature of the attacking species in the slow step.

(2)

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(b) Deduce the rate equation for the reaction using the mechanism in (a)(i).

(1)

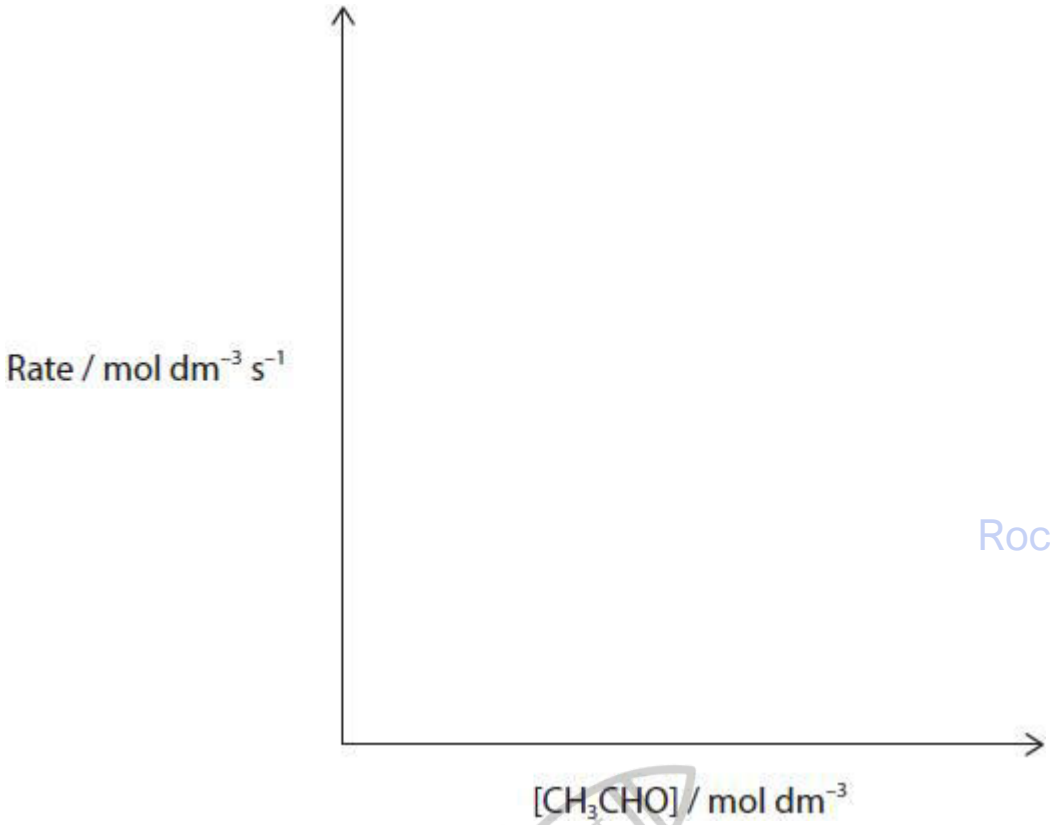
(c) Sketch a line on the axes to show how the concentration of ethanal affects the rate of this reaction.

(1)

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*(d) Carbonyl compounds can be identified from their 2,4-dinitrophenylhydrazine derivatives.

Describe this process, including in your answer

- observations you would see in the formation of the 2,4-dinitrophenylhydrazine derivative
- an outline of the steps used to separate and purify the derivative, giving the purpose of each step
- how the derivative is used to identify the original carbonyl compound.

Detailed descriptions of practical techniques are not required.

(6)

(Total for question = 14 marks)

Q19.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) The mass spectrum of compound **X** shows a large peak at m/z = 59.

This peak is due to the fragmentation of the molecular ion.
Which compound is most likely to be **X**?

(1)

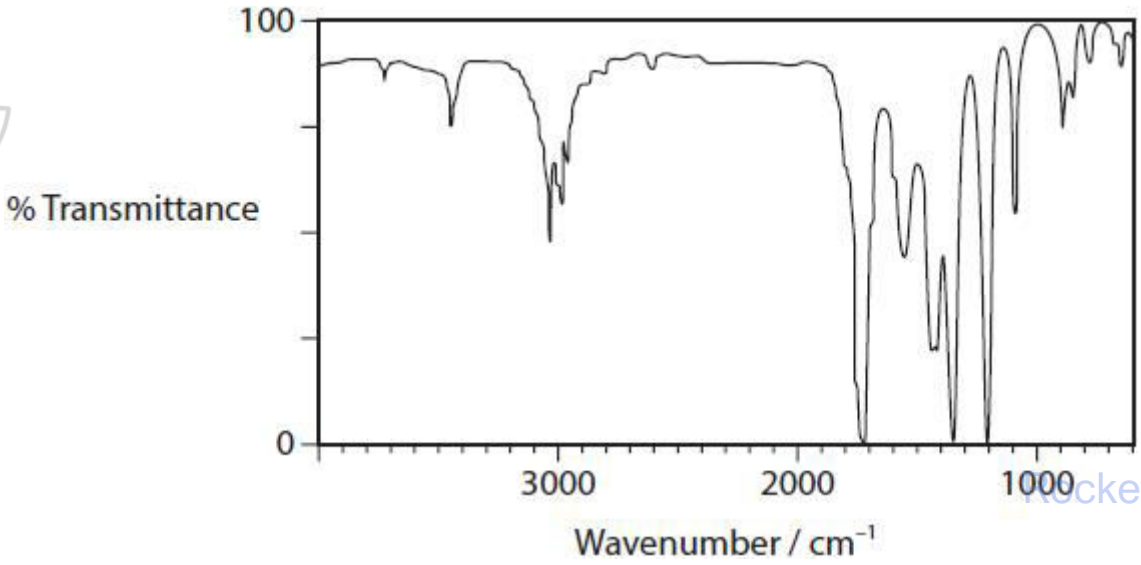
- ☒ **A** 2-methylpropan-2-ol
- ☒ **B** pentane
- ☒ **C** propanal
- ☒ **D** propanone

(b) Compound **Y** does not react with acidified aqueous sodium dichromate.

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The infrared spectrum of **Y** is shown.



Which compound is most likely to be **Y**? Refer to the Data Booklet.

(1)

- ☒ **A** 2-methylpropan-2-ol
- ☒ **B** pentane
- ☒ **C** propanal
- ☒ **D** propanone

(Total for question = 2 marks)

Q20.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Propanone reacts with HCN in the presence of KCN.

Which of these statements is **incorrect**?

- ☒ **A** CN^- acts as a nucleophile in the reaction
- ☒ **B** a racemic mixture is formed
- ☒ **C** the product is **not** optically active
- ☒ **D** the product is 2-hydroxy-2-methylpropanenitrile

(Total for question = 1 mark)

Q21.

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This question is about some reactions of propanal, CH_3CH_2CHO, and propanone, CH_3COCH_3.

(a) Propanal and propanone react similarly with lithium tetrahydridoaluminate(III).

Complete the table about these reactions.

(4)

Formula of lithium tetrahydridoaluminate(III)	
Essential reaction conditions	
Type of reaction	
Name of organic product with propanal	
Name of organic product with propanone	

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(b) State what would be **seen** when separate samples of propanal and propanone are warmed with Tollens' reagent.

(2)

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(c) When propanone reacts with iodine in the presence of aqueous sodium hydroxide, a yellow precipitate is observed.

Write an equation for this reaction.
Include state symbols.

(3)

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(d) Propanal and propanone both react with 2,4-dinitrophenylhydrazine.

The products of these reactions may be used to distinguish between separate samples of propanal and propanone.
Describe, in outline, the laboratory procedure for doing this.

(3)

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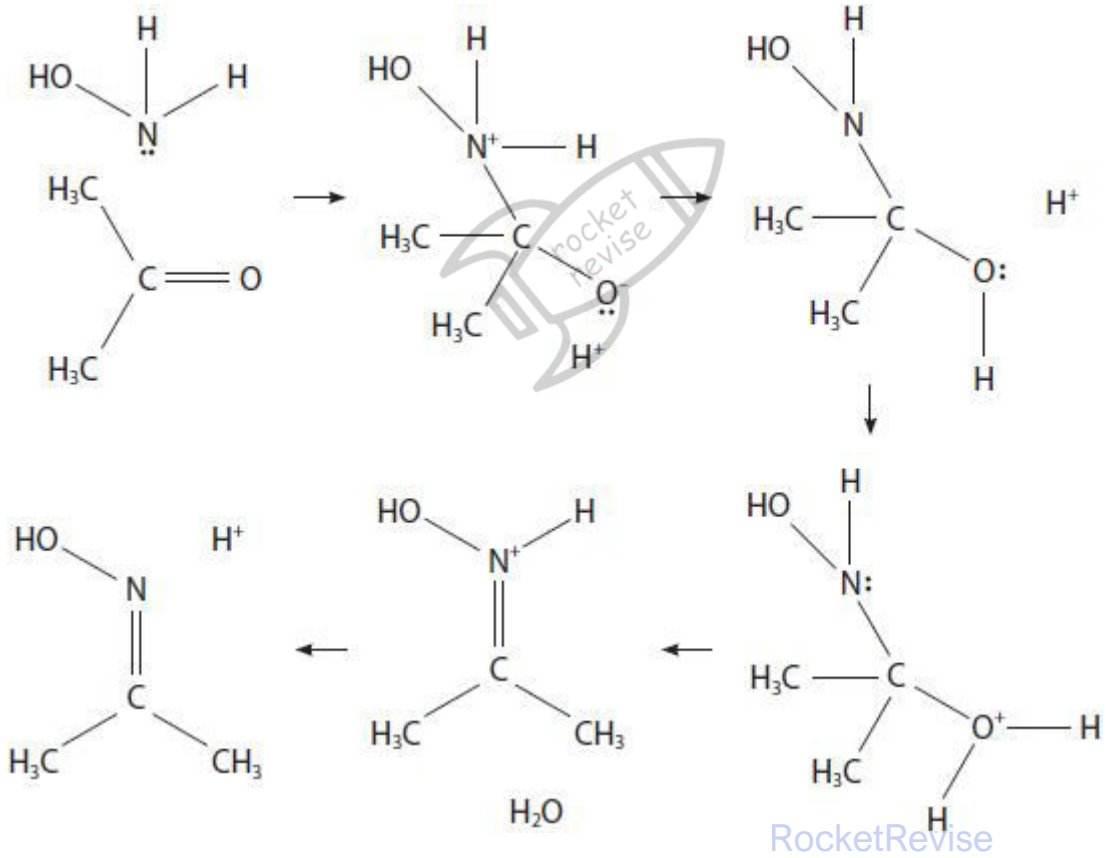
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(e) Propanone, CH_3COCH_3, reacts with hydroxylamine, NH_2OH, under mildly acidic conditions.

The first stage of the reaction is a nucleophilic addition.
Add curly arrows to complete the mechanism for this reaction.

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(4)



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(Total for question = 16 marks)

Q22.

This question is about pentan-2-one, $CH_3COCH_2CH_2CH_3$.

(a) Explain why pentan-2-one has a lower boiling temperature than pentan-2-ol.

(2)

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(b) Give a chemical test which distinguishes between pentan-2-one and pentan-3-one, $CH_3CH_2COCH_2CH_3$.

Include the results for both substances.

(2)

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(c) Give a different chemical test which distinguishes between pentan-2-one and pentanal, $CH_3CH_2CH_2CH_2CHO$.

Include the results for both substances.

(2)

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(d) Pentan-2-one, pentan-3-one and pentanal react in a similar way with one test reagent to form solid products.

Identify the reagent and describe how these solid products may be used to distinguish the original compounds.

(3)

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(e) Pentan-2-one and pentan-3-one both react with hydrogen cyanide in the presence of KCN. With pentan-2-one a racemic mixture is formed but with pentan-3-one there is only one product.

Explain this difference by referring **both** to the reaction mechanism and to the structures of the two molecules.

(4)

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(Total for question = 13 marks)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about carbonyl compounds.

(a) Propanone reacts with hydrogen cyanide, HCN, in the presence of potassium cyanide, KCN.

How is this reaction classified?

(1)

- ☐ **A** electrophilic addition
- ☐ **B** electrophilic substitution
- ☐ **C** nucleophilic addition
- ☐ **D** nucleophilic substitution

(b) What is the first step in the mechanism of the reaction between propanone and a mixture of HCN and KCN?

(1)

- ☐ **A** the lone pair of electrons on the carbon of CN^- attacking the carbonyl $C^{\delta+}$ of propanone
- ☐ **B** the lone pair of electrons on the nitrogen of CN^- attacking the carbonyl $C^{\delta+}$ of propanone
- ☐ **C** the lone pair of electrons on the oxygen of propanone attacking the $C^{\delta+}$ of HCN
- ☐ **D** the lone pair of electrons on the oxygen of propanone attacking the $H^{\delta+}$ of HCN

(c) What is the name of the organic product formed in part (a)?

(1)

- ☐ **A** 2-hydroxy-2-methylpropylamine
- ☐ **B** 2-hydroxy-2-methylethylamine
- ☐ **C** 2-hydroxy-2-methylethanenitrile

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☒ **D** 2-hydroxy-2-methylpropanenitrile

(d) Which of these carbonyl compounds would produce pale yellow crystals when warmed with iodine in the presence of sodium hydroxide?

(1)

☒ **A** butanone
☒ **B** butanal
☒ **C** pentan-3-one
☒ **D** 2-methylpropanal

(e) What is the formula of the pale yellow crystals formed in part (d)?

(1)

☒ **A** CI_4
☒ **B** CHI_3
☒ **C** CH_2I_2
☒ **D** CH_3I

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(Total for question = 5 marks)

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